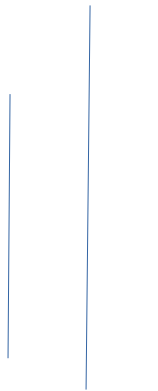


Lab Report of Computer Network (CSC 263)



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Table of Contents

1. Understanding the Network Equipment, Wiring in Details (CAT6 UTP EIA/TIA 568A/B Straight and Cross-Over Wiring and Testing).....	4
a. To understand different network equipment:	4
b. To understand Color Coding Standard of UTP Cable:	5
c. To prepare straight-through and cross-over cables and Test it:.....	5
2. Practice on Basic Networking Commands (ifconfig/ipconfig, tcpdump, netstat, ip, hostname, route, tracer/traceroute)	8
Commands and Descriptions	8
3. Overview of IP Addressing and Subnetting.....	11
Objective	11
Terminologies	11
IPv4 Addressing	11
Subnetting.....	12
IP Address Classes & Network/Host Portion:	12
Private & Special Addresses.....	13
Subnetting Overview.....	13
Subnetting Examples.....	13
Example 1 – Class C Network.....	13
Default Subnet Mask and Addresses:	14
Conclusion:	14
4. Introduction to Packet Tracer: Basic Router Configuration	15
Objective	15
Overview	15
Basic Router Commands:	15
Step 1: Open Packet Tracer and Add Devices.....	16
Step 2: Configure Router Interfaces	16
Step 3: Configure PC	17
Step 4: Test Connectivity	17
Step 5: Save Configuration.....	17
Conclusion.....	17
5. Static Routing Implementation	18
Objective	18
Introduction to Static Routing	18
Key Features of Static Routing	18
Advantages of Static Routing	18
Drawbacks of Static Routing	18
Basic Static Routing Example in Packetracer	18
6. VLAN and VLAN Trunking.....	20
Objective	20
Introduction	20
Key Concepts	20
Topology Setup:	20
Step-by-Step Configuration	20
1. Assign IP Addresses to PCs	20
2. Create VLANs on Switch 1 (S1)	20
3. Assign Ports to VLANs on Switch 1.....	21

4. Create VLANs on Switch 2 (S2)	21
5. Assign Ports to VLANs on Switch 2.....	21
6. Configure Trunk Between Switch 1 and Switch 2.....	22
7. Implementation of RIP, EIGRP, OSPF and BGP	23
Objective(s):.....	23
Background:	23
Types of Routing Protocols:	23
Router Configuration Steps (Packet Tracer Example):	23
1. Interface Configuration	23
2. RIP Configuration	24
3. EIGRP Configuration	24
4. OSPF Configuration.....	24
5. BGP Configuration.....	24
Verification Commands:.....	25
Conclusion:	25
8. DHCP, WEB, DNS, EMAIL, FTP Server Configuration.....	26
Objectives:	26
Terminologies:.....	26
a. DHCP:.....	26
Components of DHCP	26
b. Web Server	27
c. DNS.....	27
How DNS Works:.....	28
d. Mail Server.....	28
What is FTP Useful For?	29

1. Understanding the Network Equipment, Wiring in Details (CAT6 UTP EIA/TIA 568A/B Straight and Cross-Over Wiring and Testing)

Objective:

- To study different network equipment used in LAN.
- To understand Color Coding Standard of UTP Cable
- To prepare straight-through and cross-over cables and Test it

Appartus Required: Crimper, RJ 45 Jack/Plug, LAN Cable Tester, UTP, Fiber Cable, Switch/Hub/Router

a. To understand different network equipment:

1. UTP (Unshielded Twisted Pair):

Most common LAN cable with 4 twisted pairs (8 wires).

Used with RJ-45 connectors, supports up to 10 Gbps (CAT6).

2. Fiber Optic Cable:

Transmits data as light signals for high speed and long distance.

Immune to interference, used in backbone and WAN networks.

3. Router:

Connects different networks (LAN to Internet).

Works on IP addresses and provides routing functions.

4. Switch:

Connects multiple devices within a LAN.

Forwards data using MAC addresses, faster than hubs.

5. Hub:

Simple device that broadcasts data to all ports.

Causes collisions, slower and less secure than switches.

6. Access Point (AP):

Provides wireless (Wi-Fi) access to a wired LAN.

Acts as a bridge between wired and wireless networks.

7. Patch Panel:

Organizes and terminates network cables in racks.

Used for structured cabling in LAN installations.

8. Firewall:

Security device that filters network traffic.

Protects LAN from unauthorized access and attacks.

b. To understand Color Coding Standard of UTP Cable:

UTP cables follow the **EIA/TIA 568A and 568B standards** for proper wiring and termination in RJ-45 connectors. These standards ensure compatibility and reliable communication.

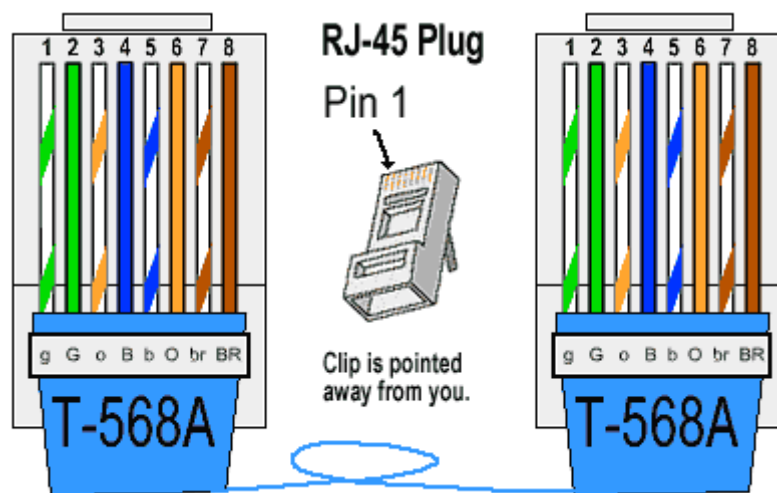


Fig: Color Coding Standard of UTP

NOTE:

In **Straight-through cable**, both ends use the same standard (A–A or B–B).

In **Cross-over cable**, one end uses **568A** and the other end uses **568B**.

Use Cases of 568A and 568B

568A Standard:

- Mostly used in **residential wiring** and government projects.
- Ensures backward compatibility with older telephone wiring.

568B Standard:

- Commonly used in **commercial networks** and data centers.
- Preferred in most Ethernet cabling systems worldwide.

c. To prepare straight-through and cross-over cables and Test it:



Here in Figure: Crimper, UTP Cable, LAN Tester, RJ 45 Jack

Step by Step Implementation:

Step 1: Strip the Cable about 1.5 or 2 inch

Step 2: Make the Pair according to Color Coding Standard

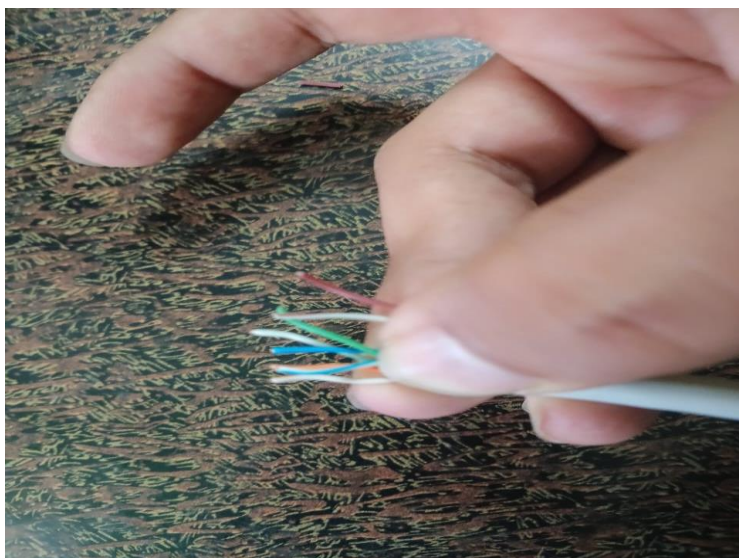
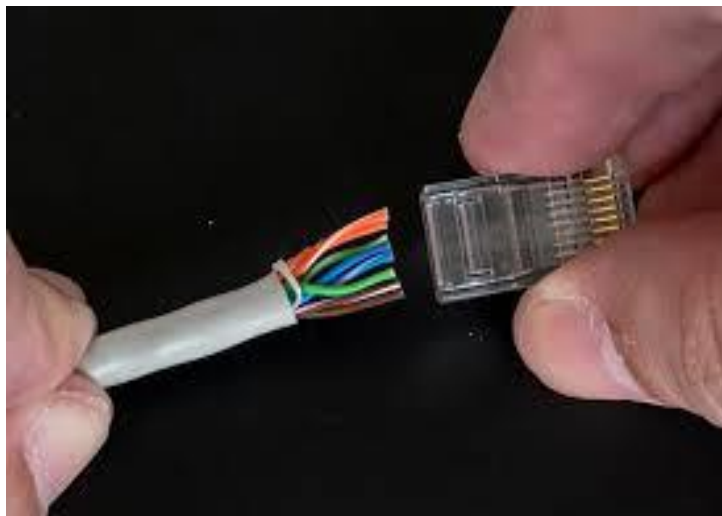


Fig: Making Pairs after stripping Cable

Step 3: Arrange and Insert the pairs in RJ45 Jack



Step 4: Make same in another side of Cable (A-A) if you want to connect different devices i.e. PC to Router, or make (A-B) if you want to connect to Similar devices

A-A → Straight Cables

A-B → Cross-Over Cables



2. Practice on Basic Networking Commands (ifconfig/ipconfig, tcpdump, netstat, ip, hostname, route, tracert/traceroute)

Objective: To understand basic command line operation with Linux and network configuration, testing and verification.

Commands and Descriptions

1. ifconfig / ipconfig

- Used to display and configure network interface settings.
- Shows IP address, MAC address, subnet mask, and status of interfaces.

```
→ ~ ifconfig

br-c8a1f9faae1b: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.18.0.1 netmask 255.255.0.0 broadcast 172.18.255.255
    ether 0a:fc:16:21:0f:91 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

br-f329a3a6905f: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.19.0.1 netmask 255.255.0.0 broadcast 172.19.255.255
    ether 06:3e:29:2a:49:d0 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 06:85:cc:68:44:97 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

t for ifconfig.

- Can manage routes, addresses, neighbors, and interfaces.

- Example: `ip addr show` displays all IP addresses.

```
→ ~ ip route show

default via 192.168.18.1 dev enp1s0 proto dhcp src 192.168.18.36 metric 100
default via 192.168.18.1 dev wlp2s0 proto dhcp src 192.168.18.7 metric 600
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
172.18.0.0/16 dev br-c8a1f9faae1b proto kernel scope link src 172.18.0.1 linkdown
172.19.0.0/16 dev br-f329a3a6905f proto kernel scope link src 172.19.0.1 linkdown
192.168.18.0/24 dev enp1s0 proto kernel scope link src 192.168.18.36 metric 100
192.168.18.0/24 dev wlp2s0 proto kernel scope link src 192.168.18.7 metric 600
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
```

3. hostname

- Displays or sets the system's hostname.
- Example: `hostname` shows the current hostname, `hostname newname` sets a new one.

```
→ ~ hostname
shawnocode
→ ~ █
```

4. netstat

- Shows active connections, listening ports, routing tables, and interface statistics.
- Example: `netstat -tulnp` shows all listening ports with processes.

```
→ ~ netstat --help
usage: netstat [-vWeenNcCF] [<Af>] -r                netstat {-V|--version|-h|--help}
       netstat [-vWnNcaeol] [<Socket> ...]
       netstat { [-vWeenNac] -i | [-cnNe] -M | -s [-6tuw] }

    -r, --route                display routing table
    -i, --interfaces           display interface table
    -g, --groups               display multicast group memberships
    -s, --statistics           display networking statistics (like SNMP)
    -M, --masquerade           display masqueraded connections

    -v, --verbose              be verbose
    -W, --wide                 don't truncate IP addresses
    -n, --numeric              don't resolve names
    --numeric-hosts            don't resolve host names
    --numeric-ports            don't resolve port names
    --numeric-users            don't resolve user names
    -N, --symbolic             resolve hardware names
    -e, --extend               display other/more information
    -p, --programs             display PID/Program name for sockets
    -o, --timers               display timers
    -c, --continuous          continuous listing

    -l, --listening            display listening server sockets
    -a, --all                  display all sockets (default: connected)
    -F, --fib                  display Forwarding Information Base (default)
    -C, --cache                display routing cache instead of FIB
    -Z, --context              display SELinux security context for sockets
```

5. route

- Displays or modifies the routing table.
- Example: **route** shows the kernel IP routing table.

```
→ ~ route
Kernel IP routing table
Destination      Gateway          Genmask          Flags Metric Ref    Use Iface
default          _gateway        0.0.0.0          UG    100    0      0 enp1s0
default          _gateway        0.0.0.0          UG    600    0      0 wlp2s0
172.17.0.0       0.0.0.0         255.255.0.0      U      0      0      0 docker0
172.18.0.0       0.0.0.0         255.255.0.0      U      0      0      0 br-c8a1f9faae1b
172.19.0.0       0.0.0.0         255.255.0.0      U      0      0      0 br-f329a3a6905f
192.168.18.0     0.0.0.0         255.255.255.0    U     100    0      0 enp1s0
192.168.18.0     0.0.0.0         255.255.255.0    U     600    0      0 wlp2s0
192.168.122.0    0.0.0.0         255.255.255.0    U      0      0      0 virbr0
```

6. traceroute / tracert

- Shows the path packets take to reach a destination.
- Helps in identifying routing problems or delays.

```
→ ~ traceroute amazon.in
traceroute to amazon.in (54.239.33.92), 30 hops max, 60 byte packets
 1 _gateway (192.168.18.1)  0.360 ms  0.310 ms  0.450 ms
 2 10.18.0.1 (10.18.0.1)  2.563 ms  3.085 ms  3.018 ms
 3 10.250.0.118 (10.250.0.118)  3.936 ms  3.963 ms  3.987 ms
 4 36.253.0.61 (36.253.0.61)  3.509 ms  3.549 ms  3.958 ms
 5 152.52.11.9 (152.52.11.9)  44.625 ms  44.789 ms  45.141 ms
 6 116.119.73.100 (116.119.73.100)  150.775 ms  149.580 ms  116.119.68.60 (116.119.68.60)  149.528 ms
```

7. tcpdump

- A packet analyzer to capture and display network traffic.
- Example: **tcpdump -i eth0** captures traffic on interface eth0.

```
→ ~ sudo tcpdump -i enp1s0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp1s0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:10:23.862490 IP shawncode.39190 > 192.168.18.33.1716: Flags [.], ack 586498342, win 502, options [nop,nop,TS val 2773582417 ecr 2850832888], length 0
14:10:23.868187 IP 192.168.18.33.1716 > shawncode.39190: Flags [.], ack 1, win 279, options [nop,nop,TS val 2850838821 ecr 2767829053], length 0
14:10:23.892046 IP shawncode.44944 > _gateway.domain: 14420+ [1au] PTR? 33.18.168.192.in-addr.arpa. (55)
14:10:23.892931 IP _gateway.domain > shawncode.44944: 14420 NXDomain 0/0/1 (55)
14:10:23.896624 IP shawncode.43272 > _gateway.domain: 59152+ [1au] PTR? 36.18.168.192.in-addr.arpa. (55)
14:10:23.897349 IP _gateway.domain > shawncode.43272: 59152 NXDomain 0/0/1 (55)
14:10:23.992191 IP shawncode.54324 > _gateway.domain: 11224+ [1au] PTR? 1.18.168.192.in-addr.arpa. (54)
14:10:23.993045 IP _gateway.domain > shawncode.54324: 11224 NXDomain 0/0/1 (54)
```

3. Overview of IP Addressing and Subnetting

Objective

- To understand the theoretical knowledge of **IPv4 addressing** and **subnetting**, including how IP addresses are structured and how networks are divided.

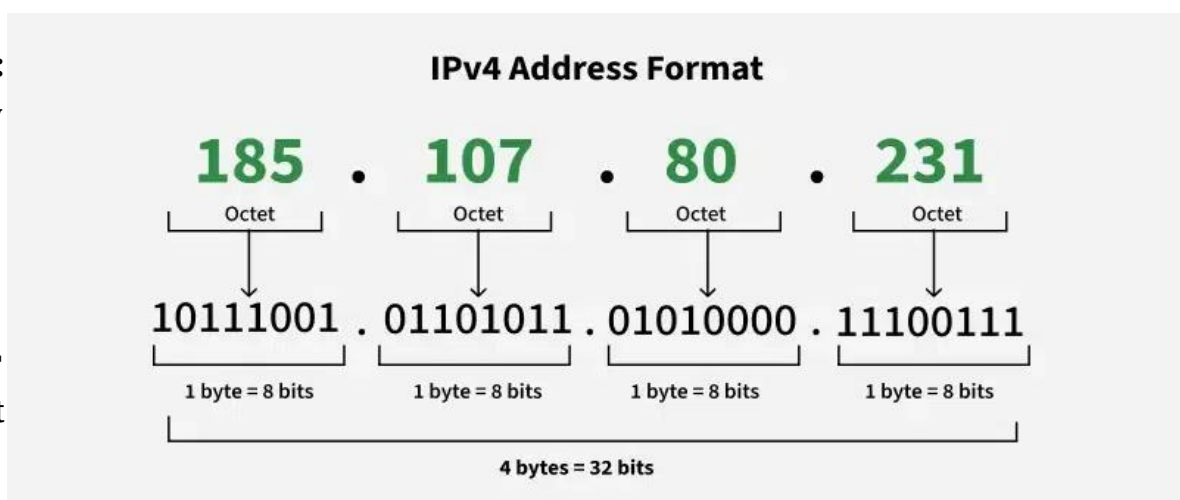
Terminologies

- **IPv4 Address:** A 32-bit number that identifies a device on a network, usually written like 192.168.1.1.
- **Host Address:** The IP of a device (computer, phone, etc.) on a network.
- **Network:** A group of devices that share the same starting part of their IP addresses.
- **Broadcast Address:** An address used to send data to all devices in a network; cannot be used by a single device.
- **Subnet:** A smaller part of a network where devices share part of the IP address.
- **Subnetting:** The process of dividing a network into smaller subnets.
- **Subnet Mask:** A number that shows which part of an IP is the network/subnet and which part is for devices.

IPv4 Addressing

- An **IPv4 address** is a 32-bit number used to identify devices on a network.
- Represented in **dotted decimal notation**: 192.168.1.10 (4 octets, 8 bits each).
- Each IP address has two parts:
 1. **Network ID** – identifies the network.
 2. **Host ID** – identifies a device on that network.

Fig:
IPv4
Address
Format



IPv4 Classes:

Class	Range	Default Subnet Mask	Usage
A	1.0.0.0 – 126.0.0.0	255.0.0.0	Large networks
B	128.0.0.0 – 191.255.0.0	255.255.0.0	Medium networks
C	192.0.0.0 – 223.255.255.0	255.255.255.0	Small networks
D	224.0.0.0 – 239.255.255.255	N/A	Multicast
E	240.0.0.0 – 255.255.255.255	N/A	Experimental

Subnetting

- Subnetting is the process of **dividing a large network into smaller subnetworks**.
- Benefits:
 1. Efficient IP address usage.
 2. Reduced network congestion.
 3. Improved security and management.

Example:

- Network: 192.168.1.0/24 → 256 addresses
- Subnet mask: 255.255.255.192 → /26 → divides network into 4 subnets:

Subnet 1: 192.168.1.0 – 192.168.1.63

Subnet 2: 192.168.1.64 – 192.168.1.127

Subnet 3: 192.168.1.128 – 192.168.1.191

Subnet 4: 192.168.1.192 – 192.168.1.255

IP Address Classes & Network/Host Portion:

Class	Range	Network Portion	Host Portion	Default Subnet Mask
A	0-127	1st octet	Last 3 octets	255.0.0.0

Class	Range	Network Portion	Host Portion	Default Subnet Mask
B	128-191	1st & 2nd octet	Last 2 octets	255.255.0.0
C	192-223	1st, 2nd, 3rd octet	Last octet	255.255.255.0
D	224-239	N/A (multicast)	N/A	N/A
E	240-255	N/A (experimental)	N/A	N/A

Network portion: Identifies the network.

Host portion: Identifies devices in the network.

Private & Special Addresses

- **Private Networks:** Not routable on the internet.
 - Class A: 10.0.0.0 – 10.255.255.255
 - Class B: 172.16.0.0 – 172.31.255.255
 - Class C: 192.168.0.0 – 192.168.255.255
- **APIPA (Auto IP):** 169.254.0.0 – 169.254.255.255
- **Special Addresses:**
 - 127.0.0.0 → Loopback (self)
 - 224.0.0.0 → Multicast
 - 255.255.255.255 → Broadcast

Subnetting Overview

- **Purpose:** Divide networks into smaller subnets for efficient IP use.
- **CIDR (Classless Inter-Domain Routing):** Lets you borrow bits from host portion to create more subnets.
- **Subnet Rules:**
 - First IP → Subnet number
 - Last IP → Broadcast address
 - Remaining → Usable hosts

Example – Class A subnet:

- Original: 10.0.0.0/8 → 16,777,214 hosts
- Borrow 1 bit → 2 subnets, 8,388,606 hosts each

Subnetting Examples

Example 1 – Class C Network

Given:

- IP Address: 192.168.1.0
- Default Subnet Mask: 255.255.255.0 (/24)
- Requirement: Divide into 4 subnets

Step 1: Determine number of bits to borrow

- Number of subnets required = 4 $\rightarrow 2^n \geq 4 \rightarrow n = 2$ bits

Step 2: New Subnet Mask

- Original mask: /24 $\rightarrow$ borrow 2 bits $\rightarrow$ /26
- Subnet mask: 255.255.255.192

Step 3: Calculate subnets

- Each subnet has $2^{(8-2)} = 64$ addresses (62 usable)
- Subnets:

Default Subnet Mask and Addresses:

IP Address	Subnet Mask	Network Address	Broadcast Address
177.100.18.4	255.255.0.0	177.100.0.0	177.100.255.255
119.18.45.0	255.0.0.0	119.0.0.0	119.255.255.255
10.10.250.1	255.0.0.0	10.0.0.0	10.255.255.255
223.23.223.109	255.255.255.0	223.23.223.0	223.23.223.255
193.100.77.8	255.255.255.0	193.100.77.0	193.100.77.255

Conclusion:

Understanding IPv4 addressing and subnetting is essential for designing networks efficiently. Subnetting helps in allocating IP addresses optimally, reducing waste, and managing large networks easily.

4. Introduction to Packet Tracer: Basic Router Configuration

Objective

- To Understand Basic Commands for Router Configuration

Overview

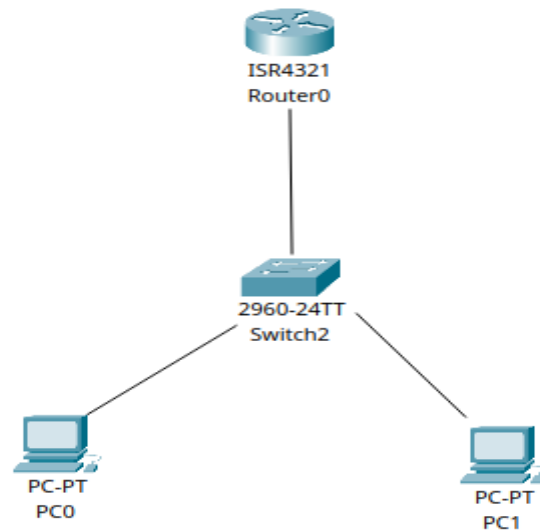
Cisco Packet Tracer is a **network simulation tool** that allows students to configure routers, switches, and other devices without physical hardware.

In this lab, we focus on **basic router configuration**, including assigning IP addresses, configuring interfaces, and verifying connectivity.

Basic Router Commands:

Command	Purpose
<code>enable</code>	Enter privileged EXEC mode
<code>configure terminal</code>	Enter global configuration mode
<code>hostname <name></code>	Set router's hostname
<code>interface <type> <number></code>	Enter interface configuration mode (e.g., interface gig0/0)
<code>ip address <IP> <Subnet></code>	Assign IP address to interface
<code>no shutdown</code>	Enable the interface (bring it up)
<code>exit</code>	Exit from current mode
<code>show running-config</code>	Display current router configuration
<code>show ip interface brief</code>	Show interface status and IP addresses
<code>ping <IP></code>	Test connectivity to another device

Setup in Packetracer



Step 1: Open Packet Tracer and Add Devices

1. Open **Cisco Packet Tracer**.
2. From the bottom-left device menu:
 - Click **Routers** → drag a router (e.g., **Generic 2811**) to the workspace.
 - Click **End Devices** → drag a PC to the workspace.
3. Connect the router and PC using a **copper straight-through cable**:
 - Click **Connections (lightning icon)** → choose **Copper Straight-Through** → click router interface `Gig0/0` → click PC NIC `FastEthernet0`.

Step 2: Configure Router Interfaces

1. Click the router → go to **CLI** tab.
2. Enter commands step by step:

Router> enable **# Enter privileged mode**

Router# configure terminal **# Enter global config mode**

```
Router(config)# hostname R1    # Set router name
R1(config)# interface gig0/0    # Enter interface configuration
R1(config-if)# ip address 192.168.1.1 255.255.255.0 # Assign IP
R1(config-if)# no shutdown     # Activate the interface
R1(config-if)# exit            # Exit interface mode
R1(config)# exit               # Exit global config
R1# show ip interface brief    # Check interface status
```

Step 3: Configure PC

1. Click the PC → **Desktop** tab → **IP Configuration**.
2. Enter:

Step 4: Test Connectivity

1. Click PC → **Desktop** → **Command Prompt**.
2. Ping the router:

Step 5: Save Configuration

- On router CLI run:

```
R1# copy running-config startup-config
```

Conclusion

- Learned **basic router commands** in Packet Tracer.
 - Configured IP addresses and verified connectivity using `ping`.
 - Understanding these basics is essential for designing and troubleshooting real networks.
-

5. Static Routing Implementation

Objective

- To understand the Static Routing, its Advantages and Drawbacks

Introduction to Static Routing

- **Routing** is the process of selecting a path for traffic in a network or between networks.
- **Static Routing** is a method where the **network administrator manually defines the routes** in the router's routing table.
- Unlike dynamic routing, static routes **do not change automatically**; the administrator must update them if the network changes.

Key Features of Static Routing

- **Manual configuration** of paths between networks.
 - Routes remain **fixed** unless manually modified.
 - Works well for **small networks** with few routers.
 - Reduces router overhead because **no routing protocols are running**.
-

Advantages of Static Routing

1. **Simple and Predictable:** Routes are fixed and easy to understand.
 2. **Low Overhead:** No CPU or memory is consumed by routing protocols.
 3. **Secure:** Less vulnerable to routing attacks since routes are manually set.
 4. **Reliable:** No dependency on dynamic protocol updates.
-

Drawbacks of Static Routing

1. **Not Scalable:** Hard to manage in large networks with many routers.
2. **Manual Updates Needed:** Any network change requires updating all affected routers.
3. **No Automatic Failover:** If a link fails, static routes do not adjust automatically.
4. **Time-Consuming:** Configuring and maintaining many static routes is tedious.

Basic Static Routing Example in Packetracer

Scenario: Two routers (R1 and R2) connected via a network.

Step 1: Configure Router Interfaces

```
R1(config)# interface gig0/0
```

```
R1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
R1(config-if)# exit
```

```
R2(config)# interface gig0/0
```

```
R2(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
R2(config-if)# no shutdown
```

```
R2(config-if)# exit
```

Step 2: Configure Static Routes

On R1:

```
R1(config)# ip route 192.168.2.0 255.255.255.0 192.168.1.2
```

On R2:

```
R2(config)# ip route 192.168.1.0 255.255.255.0 192.168.2.2
```

Step 3: Verify Connectivity

- Use `ping` from a PC on one network to a PC on the other network.
- Successful replies confirm correct static route configuration.

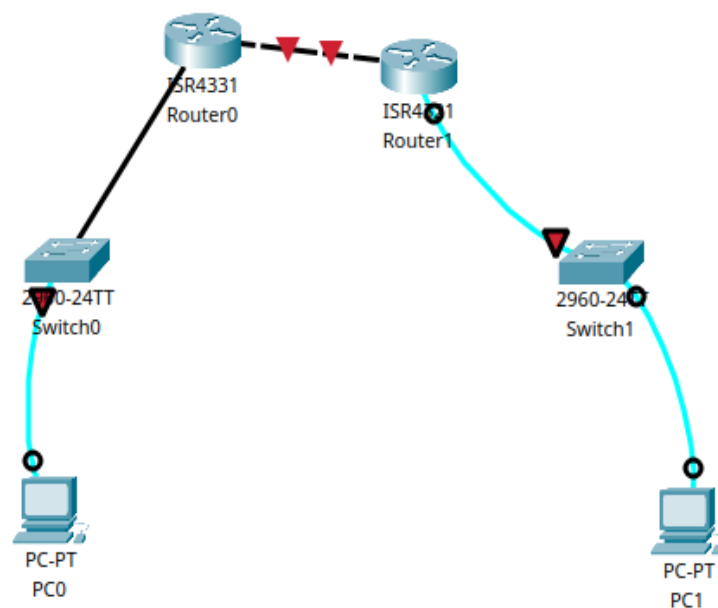


Fig: Static Routing Implementation in Packetracer

6. VLAN and VLAN Trunking

Objective

To understand LAN networking, creation of VLAN, IP addressing in the VLAN and VLAN Trunk.

Introduction

- **VLAN (Virtual Local Area Network)** is a logical subdivision of a LAN.
- VLANs allow network administrators to **segment a physical network into multiple logical networks**, improving **security and performance**.
- **VLAN Trunking** allows traffic from multiple VLANs to pass through a **single physical link** between switches.

Key Concepts

- **Access Port:** Port that belongs to a single VLAN; connects end devices like PCs.
- **Trunk Port:** Port that carries traffic for multiple VLANs; used between switches or between switch and router.
- **VLAN ID:** A unique number (1–4094) that identifies each VLAN.

Topology Setup:

- **Switch 1 (S1)** with 3 PCs (PC1, PC2, PC3).
- **Switch 2 (S2)** with 3 PCs (PC4, PC5, PC6).
- A **trunk link** connects Switch 1 and Switch 2.

We will make **two VLANs**:

- **VLAN 10 (Sales Department)** → PC1, PC2, PC4
- **VLAN 20 (HR Department)** → PC3, PC5, PC6

Step-by-Step Configuration

1. Assign IP Addresses to PCs

Example:

- **VLAN 10:** 192.168.10.1/24, 192.168.10.2/24, 192.168.10.3/24
- **VLAN 20:** 192.168.20.1/24, 192.168.20.2/24, 192.168.20.3/24

So each PC gets IP based on VLAN group.

2. Create VLANs on Switch 1 (S1)

Switch> enable

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name Sales

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name HR

Switch(config-vlan)# exit

3. Assign Ports to VLANs on Switch 1

Switch(config)# interface fa0/1

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 10

Switch(config-if)# exit

Switch(config)# interface fa0/2

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 10

Switch(config-if)# exit

Switch(config)# interface fa0/3

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 20

Switch(config-if)# exit

4. Create VLANs on Switch 2 (S2)

Switch(config)# vlan 10

Switch(config-vlan)# name Sales

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name HR

Switch(config-vlan)# exit

5. Assign Ports to VLANs on Switch 2

```
Switch(config)# interface fa0/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

```
Switch(config-if)# exit
```

```
Switch(config)# interface fa0/2
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 20
```

```
Switch(config-if)# exit
```

```
Switch(config)# interface fa0/3
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 20
```

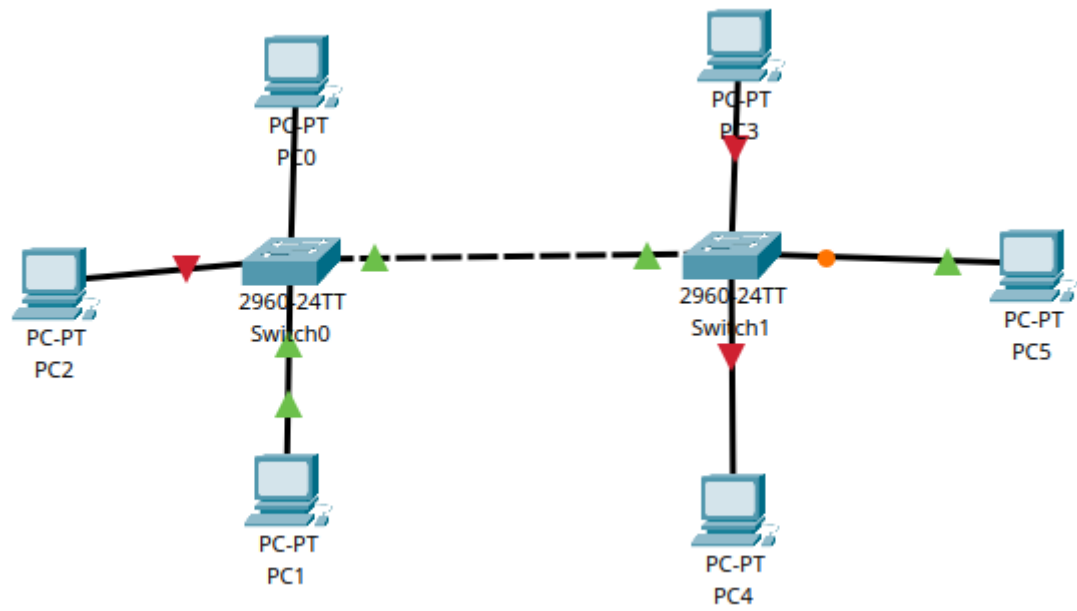
```
Switch(config-if)# exit
```

6. Configure Trunk Between Switch 1 and Switch 2

```
Switch(config)# interface fa0/24
```

```
Switch(config-if)# switchport mode trunk
```


Switch(config-if)# exit



7. Implementation of RIP, EIGRP, OSPF and BGP

Objective(s):

- To understand the basic operation of different routing protocols (RIP, EIGRP, OSPF, and BGP).
 - To configure and verify routing protocols in Packet Tracer.
 - To analyze differences between Distance Vector, Link State, and Path Vector protocols.
-

Background:

Routing protocols are sets of rules used by routers to exchange information about networks. They do not move user data directly but instead **update and maintain the routing table**, which helps routers decide the best path to reach a destination.

Types of Routing Protocols:

1. **Interior Gateway Protocols (IGP):** Work inside an Autonomous System (AS). Examples:
 - **RIP (Routing Information Protocol):** Distance vector protocol, simple, hop-count metric (max 15 hops).
 - **EIGRP (Enhanced Interior Gateway Routing Protocol):** Hybrid protocol, Cisco proprietary, uses bandwidth + delay as metrics.
 - **OSPF (Open Shortest Path First):** Link-state protocol, uses Dijkstra's SPF algorithm, very scalable and secure.
2. **Exterior Gateway Protocols (EGP):** Work between Autonomous Systems. Example:
 - **BGP (Border Gateway Protocol):** Path vector protocol, backbone of the internet, exchanges routing information between ISPs and large networks.

Router Configuration Steps (Packet Tracer Example):

1. Interface Configuration

On Router 1 (R1):

```
R1(config)#hostname R1
R1(config)#interface s0/0/0
R1(config-if)#ip address 192.168.5.1 255.255.255.0
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config)#interface fa0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
```

On Router 2 (R2) :

```
R2(config)#hostname R2
R2(config)#interface s0/0/0
R2(config-if)#ip address 192.168.5.2 255.255.255.0
R2(config-if)#no shutdown
R2(config)#interface fa0/0
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shutdown
```

2. RIP Configuration

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.5.0
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 192.168.3.0
R2(config-router)#network 192.168.5.0
```

3. EIGRP Configuration

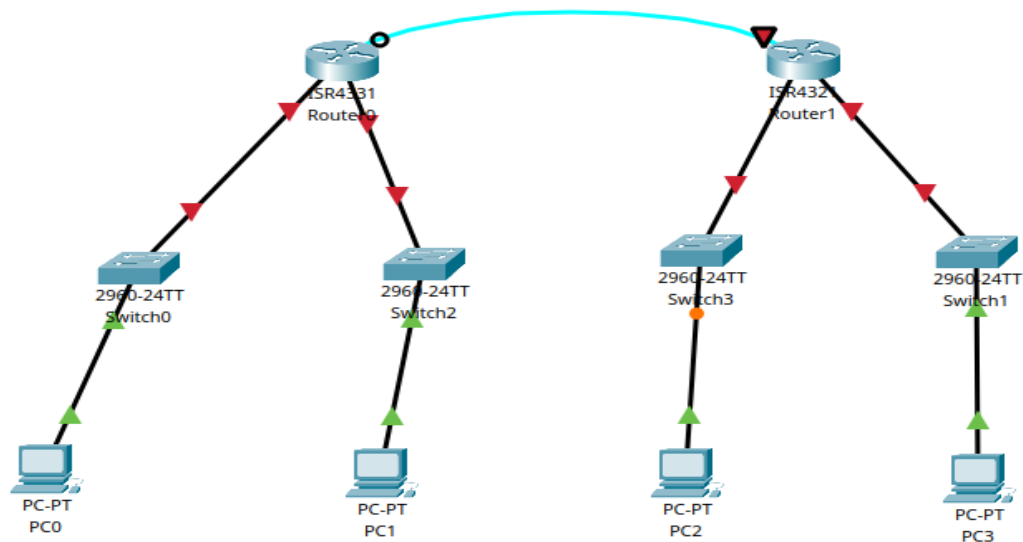
```
R1(config)#no router rip
R1(config)#router eigrp 100
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.5.0
R2(config)#no router rip
R2(config)#router eigrp 100
R2(config-router)#network 192.168.3.0
R2(config-router)#network 192.168.5.0
```

4. OSPF Configuration

```
R1(config)#no router eigrp 100
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.5.0 0.0.0.255 area 0
R2(config)#no router eigrp 100
R2(config)#router ospf 1
R2(config-router)#network 192.168.3.0 0.0.0.255 area 0
R2(config-router)#network 192.168.5.0 0.0.0.255 area 0
```

5. BGP Configuration

```
R1(config)#router bgp 100
R1(config-router)#network 192.168.1.0 mask 255.255.255.0
R1(config-router)#neighbor 192.168.5.2 remote-as 200
R2(config)#router bgp 200
R2(config-router)#network 192.168.3.0 mask 255.255.255.0
R2(config-router)#neighbor 192.168.5.1 remote-as 100
```



Verification Commands:

- `ping <IP>` → check connectivity
- `show ip route` → check routing table
- `show running-config` → check configurations

Conclusion:

- **RIP** is simple but only good for small networks.
- **EIGRP** is faster and efficient with multiple metrics.
- **OSPF** is scalable and widely used in enterprises.
- **BGP** is essential for routing between ISPs and across the internet.
- By configuring and testing each protocol, we understood how routers dynamically share routes and select the best path.

8. DHCP, WEB, DNS, EMAIL, FTP Server Configuration

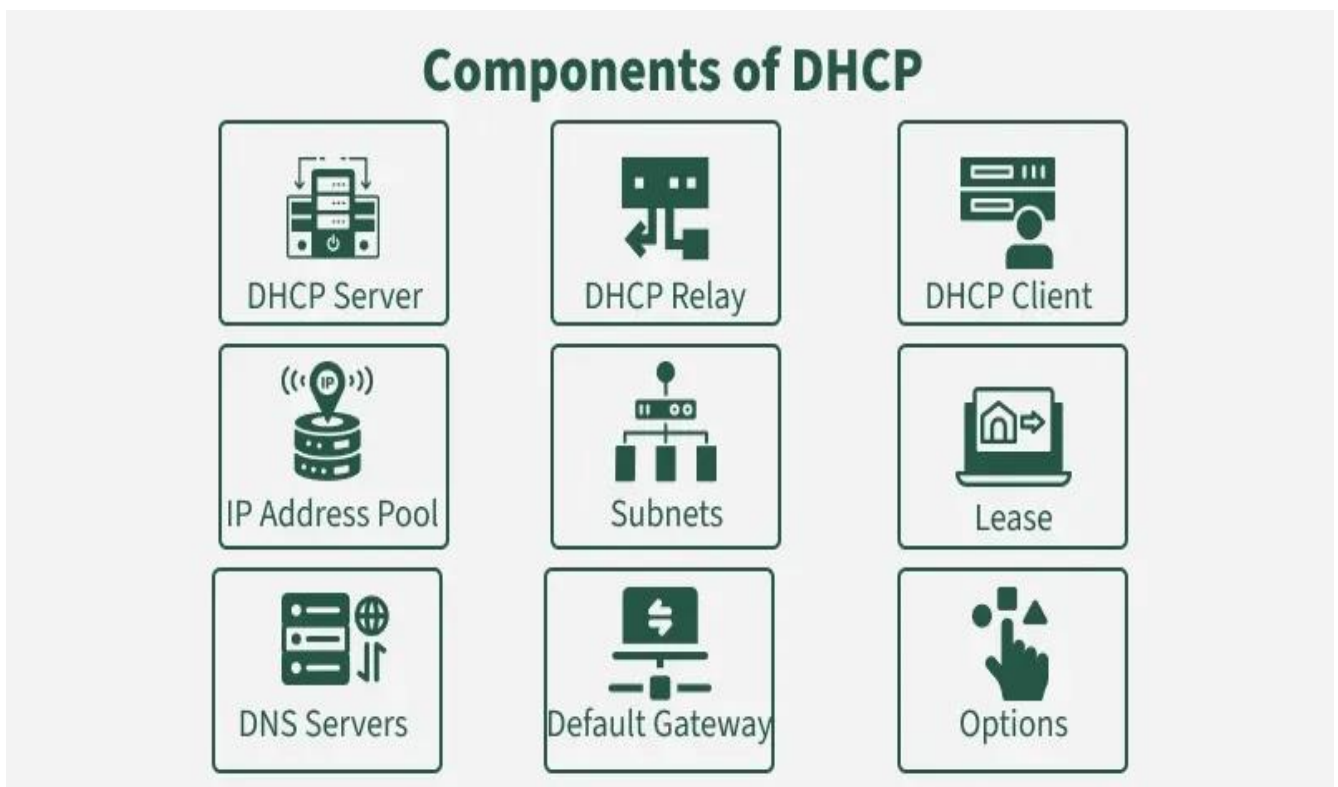
Objectives:

- To understand the working principle of **DHCP, WEB, DNS, EMAIL, and FTP Servers**.

Terminologies:

a. DHCP:

Dynamic Host Configuration Protocol is a network protocol used to automate the process of assigning IP addresses and other network configuration parameters to devices (such as computers, smartphones and printers) on a network.

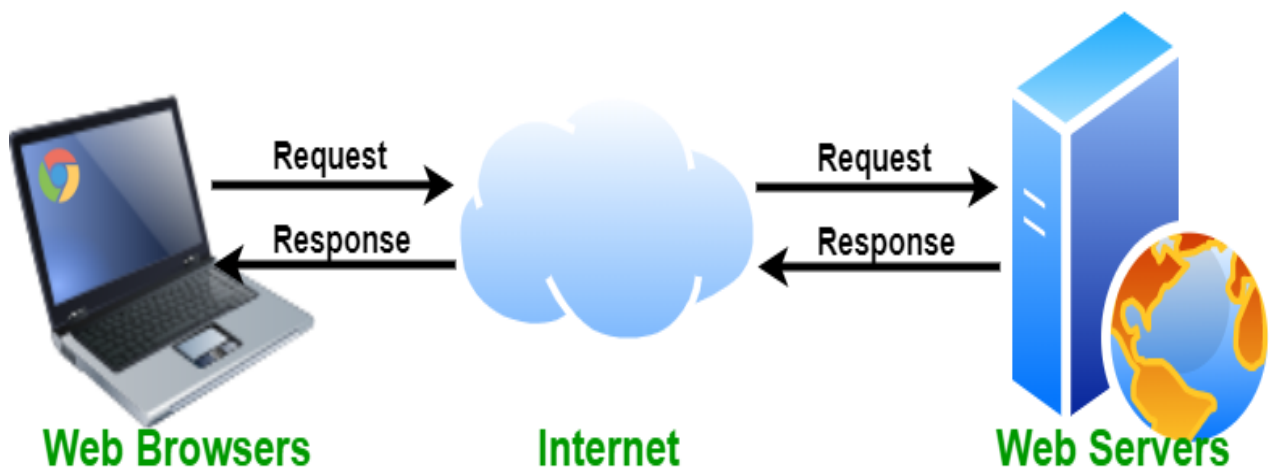


Components of DHCP

- 1. DHCP Server:** It is typically a server or a router that holds the network configuration information.
- 2. DHCP Client:** It is the endpoint that gets the configuration information from the server like any computer or mobile.
- 3. DHCP Relay Agent:** If you have only one DHCP Server for multiple LAN's then the DHCP relay agent present in every network will forward the DHCP request to the servers. This because the DHCP packets cannot travel across the router. Hence, the relay agent is required so that DHCP servers can handle the request from all the networks.
- 4. IP address pool:** It contains the list of IP address which are available for assignment to the client.
- 5. Subnet Mask:** It tells the host that in which network it is currently present.
- 6. Lease Time:** It is the amount of time for which the IP address is available to the client. After this time the client must renew the IP address.
- 7. Gateway Address:** The gateway address lets the host know where the gateway is to connect to the internet.

b. Web Server

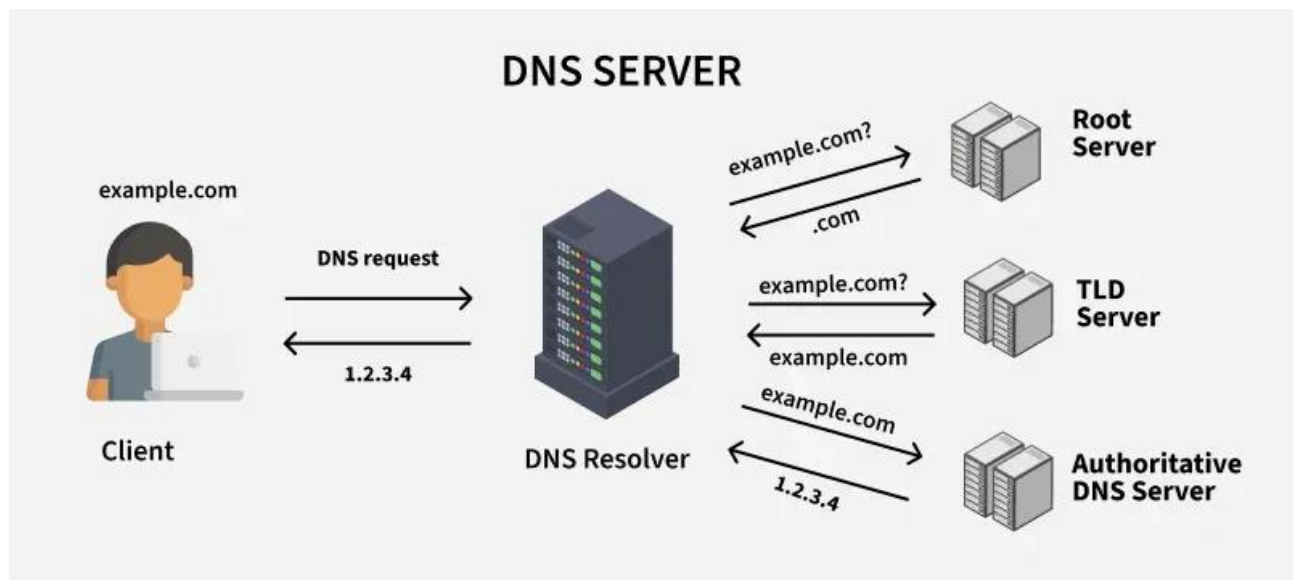
A web server is a system either software, hardware, or both that stores, processes, and delivers web content to users over the Internet using the HTTP or HTTPS protocol. When a user's browser sends a request (like visiting a website), the web server responds by delivering the appropriate resources, such as HTML pages, images, videos, or data.



DG

c. DNS

The domain name system (DNS) is a naming database in which internet [domain](#) names are located and translated into [Internet Protocol \(IP\) addresses](#). The DNS maps the name people use to locate a website to the IP address that a computer uses to locate that website.



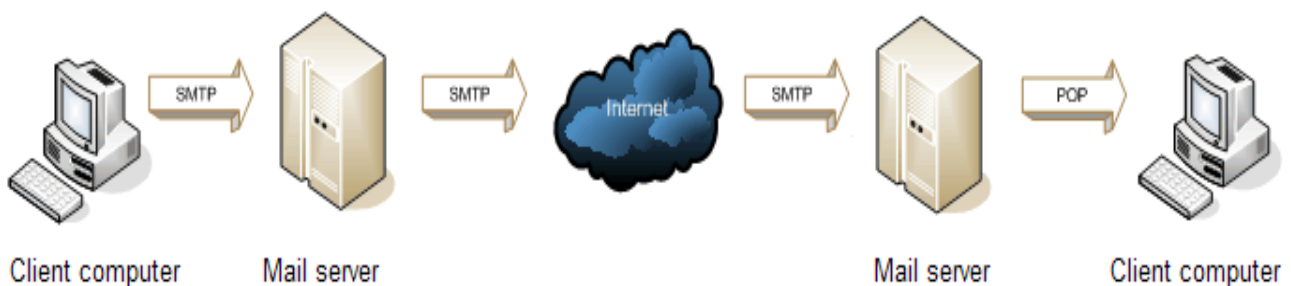
How DNS Works:

1. **User Input:** You enter a website address (for example, www.amazon.com) into your web browser.
2. **Local Cache Check:** Your browser first checks its local cache to see if it has recently looked up the domain. If it finds the corresponding IP address, it uses that directly without querying external servers.
3. **DNS Resolver Query:** If the IP address isn't in the local cache, your computer sends a request to a DNS resolver. The resolver is typically provided by your Internet Service Provider (ISP) or your network settings.
4. **Root DNS Server:** The resolver sends the request to a root [DNS server](#). The root server doesn't know the exact IP address for www.geeksforgeeks.org but knows which Top-Level Domain (TLD) server to query based on the domain's extension (e.g., .org).
5. **TLD Server:** The TLD server for .org directs the resolver to the authoritative DNS server for geeksforgeeks.org.
6. **Authoritative DNS Server:** This server holds the actual DNS records for geeksforgeeks.org, including the IP address of the website's server. It sends this IP address back to the resolver.
7. **Final Response:** The DNS resolver sends the IP address to your computer, allowing it to connect to the website's server and load the page.

d. Mail Server

A mail server (sometimes called an email server) is a software program that sends and receives email. Often, it is used as a blanket term for both mail transfer agents (MTA) and mail delivery agents (MDA), each of which perform a slightly different function.

Mail servers play a crucial role in the email delivery process. Without them, users would have no way of transferring those messages to and from other mail clients.



SMTP, POP3 and IMAP4

1. SMTP (Simple Mail Transfer Protocol) is a protocol that is used when e-mails are delivered from clients to servers and from servers to other servers. It uses port 25, 2525 and 465.
2. POP3 (Post Office Protocol version 3) is used to retrieve email from the servers. It uses port 110 and 995.
3. IMAP4 (Internet Message Access Protocol Version 4) is a further development of the POP3 protocol and is used to read e-mail from mail servers. IMAP4 is not used as much as POP3, but many modern mail servers have support for IMAP4. It uses port 143 and 993.

File Transfer Protocol (FTP):

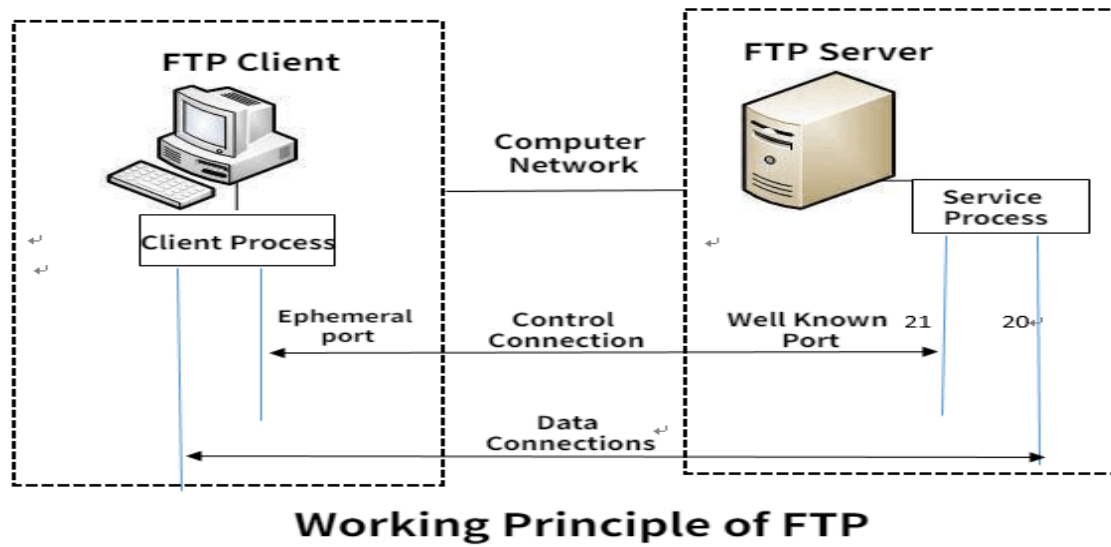
File transfer protocol (FTP) is a method for moving files between computers on a network. It's a set of instructions that computers use to communicate and exchange files. You can use FTP to upload files to a server, download files from a server, or transfer data between computers on a local network.

What is FTP Useful For?

FTP is especially useful for:

- **Transferring Large Files:** FTP can transfer large files in one shot; thus applicable when hosting websites, backing up servers, or sharing files in large quantities.
- **Remote File Management:** Files on a remote server can be uploaded, downloaded, deleted, renamed, and copied according to the users' choices.
- **Automating File Transfers:** FTP is a great protocol for the execution of file transfers on predefined scripts and employments.

- **Accessing Public Files:** Anonymous FTP means that everybody irrespective of the identity is allowed to download some files with no permissions needed.



Lab Report of Computer Network (CSC 263)



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